

# Submission Summary

**Conference Name**

Workshop on Engineering Applications 2026

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**Track Name**

Simulation Systems

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**Paper ID**

83

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**Paper Title**

Optimal Control Methods for Compensating Packet Losses in Networked Control Systems over Markov Channels

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**Abstract**

With the advancement of the Internet of Things (IoT), it is desirable to develop control systems that include communication loops in feedback loops. For example, wireless networked control systems are particularly relevant in robotic assistance and rehabilitation systems, precision agriculture, space exploration, and cooperative robotics, among others. However, wireless networked control systems are sensitive to communication problems such as delays, packet loss, and bandwidth limitations. Among these, packet loss can degrade the performance of the closed-loop system and affect its stability. Therefore, this paper introduces optimal control methods to compensate for packet loss in a communication channel. The channel is modeled as a finite-state Markov chain that captures the stochastic nature of packet loss. Two controllers are tested: Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC), both aimed at reducing losses and managing control effort. The choice of these two techniques is based on their shared characteristics of optimizability and closed-loop robustness. However, our proposal focuses on the predictive control action capability offered by MPC, which allows a set of control actions to be sent in each packet, enabling the controller to mitigate the effects of packet loss. Simulations were performed to observe the effect of lost packages on the control action and state variables of a dynamical system, using RMS values to quantify performance. The results show that LQR is sensitive to packet loss, even decreasing control effort and resulting in degraded speed response. MPC, on the other hand, maintains stability by gradually increasing control actions, demonstrating a better trade-off between performance and control effort. These findings suggest that predictive control is more robust to typical network uncertainties, at the cost of increasing control actions, but providing consistent performance while managing control energy more effectively.

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**Submission Files**Optimal Control Methods for Compensating packet Losses.pdf (3.1 Mb, 2026-05-09 18:53:33 GMT-5)

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